

Ryan Ni

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EDUCATION

University of California San Diego – UCSD

Expected Grad: **June 2027**

B.S. in **Computer Science**, B.A. **Media Arts (Cinematic/Film Studies)**

3.8 GPA

Courses: Software Engineering, Operating Systems, Databases, AI/ML, Parallel Computing, Networks

18x Hackathon Winner: Top 5 Finalist @ OpenAI HQ Hackathon, #2 Globally @ \$3M pump.fun Build in Public Hackathon, Top 5/1,500+ interns @ Apple 2025 Hackathon/iContest, #1 Best Project @ UIUC, Texas A&M, UCLA, UCSD, UCSB, UCSC

EXPERIENCE

Apple | **Software Engineer Intern – Intelligent System Experience (ISE)**

Cupertino, CA | Jun 2025 – Sep 2025

- Placed **Top 5/1,500+ interns** for Best Future Apple Feature at **iContest 2025** and pitched feature to **7 Executives**; new feature is being shipped to production on **iOS/macOS**
- Refactored **Apple Intelligence (AI)** Notification Settings by streamlining an **Objective-C** data provider, **reducing code duplication 66%** and resolving system/UI rerendering issues
- Architected **2 new AI-powered iMessage features** using **data detectors**, **message vectorization**, and a redesigned Contacts card UI for pin suggestions, pin summaries, and semantic message search
- Built end-to-end **Swift UI** notification testing app to debug daemons, entitlements, system bundles, notification UI

Q Lab | **AI/ML Research (Research Project Lead, TRELS candidate)**, under Dr. Qin

San Diego, CA | May 2025 – Present

- Investigating long-horizon strategic decision-making by developing a **Gymnasium-based RL environment in Python/NumPy** for seeded Polytopia weekly challenges and enabling 64 parallel agent rollouts
- Studying** reward design and long-horizon credit assignment by comparing **8+ reward formulations across 50+ experiments**, using Weights & Biases to track training dynamics and final challenge performance
- Developing strategic policies in PyTorch through behavioral cloning on **100+ expert state-action trajectories** scraped from YouTube and ranked replays, initializing policies for downstream RL and search

XR Lab | **Software Engineer & Research Assistant**, under Dr. Kirsh

San Diego, CA | Jan 2025 – Jun 2025

- Developed a **PyTorch** speech emotion recognition (SER) model achieving **80% precision** using Kaggle/EmoDB datasets
- Improved ML data-pipeline throughput **~40%** by migrating experimental data from **CSV/PostgreSQL** to **HDF5**, enabling near real-time training and evaluation

SDxUCSD | **Events Director**

San Diego, CA | Nov 2024 – Present

- Founded a **300+ member** technical founder community, organizing **20+ events** with **YC, Anthropic, xAI, Vercel, etc.**

PROJECTS / LEADERSHIP

Orca | **Founder – Lightyear (HF0 incubated)** funded startup

San Francisco, CA | Summer 2026

- Built a consumer agent and a **NATS JetStream** distributed AI agent orchestration network for asynchronous, non-blocking task dispatch to coordinate **1,000s of agents** across independently operated nodes
- Built a task marketplace enabling workloads including **data labeling, world-model generation, and distributed AI operations** to be decomposed and executed across an agent cluster; federated learning, engagement hacking, etc.
- Secured **22 contracts**, led a **5-person team**, and **scaled installations from 0 to 100+** colleges before demo day

YOLO dex – YOLO labeling Agent | **Top 5 Finalist @ OpenAI Hackathon – Codex**

[\[Demo\]](#)

- Reduced manual video-labeling workflows from **~10 hours to ~10 minutes (60x)**, with the approach receiving validation from **Sam Altman** during judging; [\[Me Pitching to Sam Altman\]](#)
- Engineered an **RL fine-tuning (RLFT)** pipeline for specializing **YOLO classifiers** across granular labeling requirements
- Automated data end-to-end pipeline: YouTube URLs → frame extraction via sub-agents → annotated datasets

ClawCraft (Opal) – AI Agents for Gaming and Data | **#2 @ \$3M pump.fun Global Hackathon**

[\[Pump.fun X Post\]](#)

- Built AI agent harness enabling OpenClaw nodes to autonomously **play Minecraft** and **stream on Twitch**, creating an open-world sandbox to studying multi-agent coordination and emergent behavior at scale
- Worked on an imitation learning pipeline that converts user telemetries into for train and evaluate agent gameplay

SKILLS

AI/ML: PyTorch, JAX, TensorFlow, OpenCV, Scikit-Learn, ONNX, LangChain, Reinforcement Learning, MCP, MediaPipe, Roboflow

Systems & Infrastructure: Linux, Docker, Kubernetes, CUDA, ROCm, AWS, GCP, FastAPI, Nginx, FFmpeg, Git, Cargo, uv/pip, Ruff

Languages: Python, C++, C, Rust, Go, TypeScript/JavaScript, Java, Swift, Objective-C, SQL, R, Dart

Data & Web: PostgreSQL, HDF5, MongoDB, Supabase, Firebase, React/Next.js, Tailwind, Vercel